

1. Get 3D scan of campus

Scaniverse (mobile app): free and easy to use but after my test scans I don't think this is viable for our map if we want it to look like the campus. The scans lack a lot of detail and there are weird gaps, inconsistencies, and warps. May still be an option for scanning teachers when making player models but idk.

Polycam: Another mobile app but requires subscription for use. I haven't tested this yet because it requires payment but it looks like it is better than scaniverse.

https://www.youtube.com/watch?v=Ibcl_vGJj_I (after watching this video this app definitely is a decent upgrade from scaniverse but it can't achieve the level of detail that a more pricey scanner can)

Microsoft 360 scanner kinect infrared laptop software: Scanning method that Mr. Theiss mentioned. <https://www.youtube.com/watch?v=JluBO5ZxAQ8> Big increase in accuracy compared to the mobile apps but seemingly a lot more effort/work required. The video also doesn't show what a much larger scan would look like for rooms/buildings so I don't know if the quality of the scan would change. Software installations and Xbox kinect needed.

Expensive Scanner: [FJ Dynamics Trion V4e LiDAR Kit Potable LiDAR Scanner FJDV4E](#) Video demo on this website showcases a use that seems very similar to what we are trying to achieve and it can probably accomplish this with a much higher level of accuracy but it costs \$4,000 which doesn't seem too realistic budget wise. There are other scanners out there but most that can do large scans for buildings and rooms with detail weren't cheaper than this one (at least from what I saw)

Drone Scanning: Mentioned but don't know too much about this. I think we said this was cheaper than a scanner like the one above though.

Photogrammetry: One that I see pops up that involves using a ton of photos rather than a scan to create a model but I don't know if this is the best for large models like we're trying to do. Seems like it could be inefficient.

Once method for scanning is determined, send the scan as a fbx file and open in blender. (fbx is what I used with the scaniverse test but other file types might work or be better I just don't know yet. I think the polycam video mentions .obj)

2. Clean up scan file in Blender

After scanning and depending on the method that was used, the raw room scans will usually have a variety of millions of triangles, holes & floating geometry, bad lighting/shadows baked in, and/or no clean walls as well as floors. This will kill performance and not look as good in Unity if imported directly. <https://www.blender.org/download/>

I have never used Blender before but based on research this is what will usually be done:

- Reduce Polygon Count (Decimate)
 - Add Decimate Modifier
 - Set ratio: 0.1 – 0.3 (start here)
 - Apply

*This alone can cut file size by 80–90%.

- Fix Geometry
 - Delete floating junk
 - Close holes
 - Straighten wall/floors
 - Remove scan artifacts
- Optional: Retopology (Advanced <https://www.youtube.com/watch?v=X2GNyEUvpD4>)

If we want a real game-quality level:

- Rebuild walls/floors with clean geometry
- Use scan only as reference

*Not required for small projects, but best practice.

<https://www.youtube.com/watch?v=f7F8jNDWae0&t=18s> (good starting point. Quick 10 minute video about clean up process)

<https://www.youtube.com/watch?v=s6OOQ0PbKsM&t=29s> (long 2hr video shows blender tools he uses for clean up)

https://www.youtube.com/watch?v=_pzTK-LBm3o (hole fixing)

3. Optimize for Gameplay in Blender

Before exporting we can add proper collisions. For simplicity it may be best to just make simple collider meshes: boxes for walls, planes for floors, and cubes for furniture or objects (more detailed colliders may also be an option but I haven't looked into it yet). After creating the colliders you need to name them. This whole optimization can be done in blender but colliders can also be added in Unity. I'll note to see how easy or hard the process is in Blender after a good test. After this is done we export for Unity as a fbx file.

<https://www.youtube.com/watch?v=Vri-8ZJ3H2M> (custom collision for Unity in Blender)

<https://www.youtube.com/watch?v=NkLFEwJbedA> (Blender Geometry and Colliders)

https://www.youtube.com/watch?v=bX_R0cQHKJg (How to Add Colliders in Unity)

4. Import into Unity

1. Drag FBX into Assets/
2. Drop into Scene
3. Scale if needed
4. Add:
 - Mesh Colliders (if missing)
 - Lighting
 - Player Controller

<https://www.youtube.com/watch?v=sZ8lvoUtGYg>